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A Tracking System Using Location Prediction an Dynamic Threshold for Minimizing SMS Deliver

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Abstract: **In this paper, a novel method called location-based delivery (LBD), which combines the short message service (SMS) and global position system (GPS), is proposed, and further, a realistic system for tracking a target's movement is developed. LBD reduces the number of short message transmissions while maintaining the location tracking accuracy within the acceptable range. The proposed approach, LBD, consists of three primary features: Short message format, location prediction, and dynamic threshold. The defined short message format is proprietary. Location prediction is performed by using the current location, moving speed, and bearing of the target to predict its next location. When the distance between the predicted location and the actual location exceeds a certain threshold, the target transmits a short message to the tracker to update its current location. The threshold is dynamically adjusted to maintain the location tracking accuracy and the number of short messages on the basis of the moving speed of the target. The experimental results show that LBD, indeed, outperforms other methods because it satisfactorily maintains the location tracking accuracy with relatively fewer messages.**

Index Terms: **Global positioning system (GPS), location tracking, mobile phones, prediction algorithms, short message service (SMS).**

I. INTRODUCTION

The global position system (GPS) has become a common functionality in handheld devices, and therefore, several location-tracking applications have been developed [1]–[14], including continuous location-tracking of elders and children for safety reasons or to prevent them from being lost [1], [2], car monitoring and tracking [3]–[5], and intelligent transportation systems [6].

The GPS is used to obtain the location information of a target (e.g., a mobile device). However, most of the above-cited works used either an 802.11 wireless network or the short message service (SMS) to transmit the location information of a target to a tracker. For example, Lee *et al*. proposed a real-time location tracking system [1] for childcare or elde plications. It transmits the location information o device to a central GPS application server throug wireless networks. This application allows the ser taneously monitor multiple targets (e.g., elders or cl ther, Choi *et al*. assumed that the location informati is transmitted through wireless networks. Their wor proposing a geolocation update scheme to decreas frequency [7]. Lita *et al*. proposed an automobile system by using SMS [3]. The proposed system, terconnected with the car alarm system, transmits owner's mobile phone in the event of a car theft (e. of the car alarm, starting of the engine) or provi tion for monitoring adolescent drivers (e.g., exceed limit or leaving a specific area). Hameed *et al*. pr monitoring and tracking system that uses both SM to prevent car theft [5]. Anderson *et al*. proposed tion information system [6]. In this system, a harc called StarBox, which is equipped with a global sy bile communications (GSM) modem and a GPS uni in a vehicle to track the vehicle's location. StarB short messages containing its GPS coordinates to 30-s intervals. The users can send short messages to determine the expected arrival time of buses at th

Although transmitting the geolocation informati *via* wireless networks is effective when both the ta tracker are within Wi-Fi coverage area, the 802.11 works are not always accessible. When the target c is unable to access Wi-Fi, it is impossible to perf tracking. Therefore, SMS is a relatively more relia ible solution because of its widespread use (i.e., we worldwide) [6], [8]. However, SMS is a user-pay objective of this study is to minimize the transmi a tracking system by minimizing the number of SM sions while maintaining the location tracking accur

In this paper, a novel method called location- ery (LBD), which combines SMS and GPS, is pr further, a realistic system to perform precise loca is developed. LBD mainly applies the following t techniques: Location prediction and dynamic thre tion prediction is performed by using the current lo ing speed, and bearing of the target to predict it tion. When the distance between the predicted loc actual location exceeds a certain threshold, the tar a short message to the tracker to update its current l dynamic threshold maintains the location tracking a number of short messages on the basis of the mov the target.

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